BIMBINGAN TUGAS AKHIR #9

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1 Reparameterization Trick dan Backpropagation pada Bobot Berdistribusi

1.1 Reparameterization Trick pada Bobot Berdistribusi

1.1.1 Tinjau Satu Bobot

Untuk setiap bobot $w_{k,j}^{(\ell)}$ pada layer $\ell \in \{1, \ldots, L\}$ ($k = 1, \ldots, n_{\ell-1}$; $j = 1, \ldots, n_\ell$) yang berdistribusi Normal dengan rata-rata $\mu_{k,j}^{(\ell)} \in \mathbb{R}$ dan variansi $\left(\sigma_{k,j}^{(\ell)}\right)^2 > 0$:

$$w_{k,j}^{(\ell)} \sim \mathcal{N}\left(\mu_{k,j}^{(\ell)}, \left(\sigma_{k,j}^{(\ell)}\right)^2\right) \tag{1}$$

Standarisasi (*reparameterization*) terhadap peubah acak Normal Baku $z_{k,j}^{(\ell)} \sim \mathcal{N}(0,1)$ dirumuskan sebagai:

$$z_{k,j}^{(\ell)} = \frac{w_{k,j}^{(\ell)} - \mu_{k,j}^{(\ell)}}{\sigma_{k,j}^{(\ell)}} \sim \mathcal{N}(0,1) \quad \Longleftrightarrow \quad w_{k,j}^{(\ell)} = \mu_{k,j}^{(\ell)} + \sigma_{k,j}^{(\ell)} z_{k,j}^{(\ell)} \tag{2}$$

dengan turunan parsial lokal terhadap parameter distribusi $\mu_{k,j}^{(\ell)}$ dan $\sigma_{k,j}^{(\ell)}$:

$$\frac{\partial w_{k,j}^{(\ell)}}{\partial \mu_{k,j}^{(\ell)}} = 1, \qquad \frac{\partial w_{k,j}^{(\ell)}}{\partial \sigma_{k,j}^{(\ell)}} = z_{k,j}^{(\ell)} \tag{3}$$

1.1.2 Tinjau Satu Neuron

Untuk neuron ke-j pada layer ℓ, vektor bobot $\mathbf{w}_j^{(\ell)} \in \mathbb{R}^{n_{\ell-1} \times 1}$ berdistribusi Normal Multivariat dengan vektor rata-rata $\boldsymbol{\mu}_j^{(\ell)} \in \mathbb{R}^{n_{\ell-1} \times 1}$ dan matriks kovariansi diagonal $\Sigma_j^{(\ell)} \in \mathbb{R}^{n_{\ell-1} \times n_{\ell-1}}$:

$$\underbrace{\begin{bmatrix} w_{1,j}^{(\ell)} \\ w_{2,j}^{(\ell)} \\ \vdots \\ w_{n_{\ell-1},j}^{(\ell)} \end{bmatrix}}_{\mathbf{w}_j^{(\ell)}} \sim \mathcal{N}\left(\underbrace{\begin{bmatrix} \mu_{1,j}^{(\ell)} \\ \mu_{2,j}^{(\ell)} \\ \vdots \\ \mu_{n_{\ell-1},j}^{(\ell)} \end{bmatrix}}_{\boldsymbol{\mu}_j^{(\ell)}}, \underbrace{\begin{bmatrix} \left(\sigma_{1,j}^{(\ell)}\right)^2 & 0 & \ldots & 0 \\ 0 & \left(\sigma_{2,j}^{(\ell)}\right)^2 & \ldots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \ldots & \left(\sigma_{n_{\ell-1},j}^{(\ell)}\right)^2 \end{bmatrix}}_{\Sigma_j^{(\ell)}} \right) \tag{4}$$

Standarisasi terhadap vektor Normal Multivariat Baku $\mathbf{z}_j^{(\ell)} = \begin{bmatrix} z_{1,j}^{(\ell)} & z_{2,j}^{(\ell)} & \ldots & z_{n_{\ell-1},j}^{(\ell)} \end{bmatrix}^T \sim \mathcal{N}(\mathbf{0}, I_{n_{\ell-1}})$:

$$\mathbf{z}_j^{(\ell)} = \left(\Sigma_j^{(\ell)}\right)^{-1/2} \left(\mathbf{w}_j^{(\ell)} - \boldsymbol{\mu}_j^{(\ell)}\right) \sim \mathcal{N}(\mathbf{0}, I_{n_{\ell-1}}) \quad \Longleftrightarrow \quad \mathbf{w}_j^{(\ell)} = \boldsymbol{\mu}_j^{(\ell)} + \left(\Sigma_j^{(\ell)}\right)^{1/2} \mathbf{z}_j^{(\ell)} \tag{5}$$

Karena $\left(\Sigma_j^{(\ell)}\right)^{1/2}$ merupakan matriks diagonal, perkalian matriks $\left(\Sigma_j^{(\ell)}\right)^{1/2} \mathbf{z}_j^{(\ell)}$ ekuivalen dengan perkalian elemen-demi-elemen ($\odot$):

$$\underbrace{\begin{bmatrix} w_{1,j}^{(\ell)} \\ w_{2,j}^{(\ell)} \\ \vdots \\ w_{n_{\ell-1},j}^{(\ell)} \end{bmatrix}}_{\mathbf{w}_j^{(\ell)}} = \underbrace{\begin{bmatrix} \mu_{1,j}^{(\ell)} \\ \mu_{2,j}^{(\ell)} \\ \vdots \\ \mu_{n_{\ell-1},j}^{(\ell)} \end{bmatrix}}_{\boldsymbol{\mu}_j^{(\ell)}} + \underbrace{\begin{bmatrix} \sigma_{1,j}^{(\ell)} & 0 & \ldots & 0 \\ 0 & \sigma_{2,j}^{(\ell)} & \ldots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \ldots & \sigma_{n_{\ell-1},j}^{(\ell)} \end{bmatrix}}_{\left(\Sigma_j^{(\ell)}\right)^{1/2}} \underbrace{\begin{bmatrix} z_{1,j}^{(\ell)} \\ z_{2,j}^{(\ell)} \\ \vdots \\ z_{n_{\ell-1},j}^{(\ell)} \end{bmatrix}}_{\mathbf{z}_j^{(\ell)}} = \begin{bmatrix} \mu_{1,j}^{(\ell)} \\ \mu_{2,j}^{(\ell)} \\ \vdots \\ \mu_{n_{\ell-1},j}^{(\ell)} \end{bmatrix} + \begin{bmatrix} \sigma_{1,j}^{(\ell)} \\ \sigma_{2,j}^{(\ell)} \\ \vdots \\ \sigma_{n_{\ell-1},j}^{(\ell)} \end{bmatrix} \odot \begin{bmatrix} z_{1,j}^{(\ell)} \\ z_{2,j}^{(\ell)} \\ \vdots \\ z_{n_{\ell-1},j}^{(\ell)} \end{bmatrix} \tag{6}$$

1.1.3 Tinjau Satu Layer

Dengan menggabungkan seluruh kolom neuron $j = 1, \ldots, n_\ell$ ke dalam matriks bobot $W^{(\ell)} \in \mathbb{R}^{n_{\ell-1} \times n_\ell}$ pada layer $\ell \in \{1, \ldots, L\}$:

$$\underbrace{\begin{bmatrix} w_{1,1}^{(\ell)} & \cdots & w_{1,n_\ell}^{(\ell)} \\ \vdots & \ddots & \vdots \\ w_{n_{\ell-1},1}^{(\ell)} & \cdots & w_{n_{\ell-1},n_\ell}^{(\ell)} \end{bmatrix}}_{W^{(\ell)}} = \underbrace{\begin{bmatrix} \mu_{1,1}^{(\ell)} & \cdots & \mu_{1,n_\ell}^{(\ell)} \\ \vdots & \ddots & \vdots \\ \mu_{n_{\ell-1},1}^{(\ell)} & \cdots & \mu_{n_{\ell-1},n_\ell}^{(\ell)} \end{bmatrix}}_{M^{(\ell)}} + \underbrace{\begin{bmatrix} \sigma_{1,1}^{(\ell)} & \cdots & \sigma_{1,n_\ell}^{(\ell)} \\ \vdots & \ddots & \vdots \\ \sigma_{n_{\ell-1},1}^{(\ell)} & \cdots & \sigma_{n_{\ell-1},n_\ell}^{(\ell)} \end{bmatrix}}_{\Sigma^{(\ell)}} \odot \underbrace{\begin{bmatrix} z_{1,1}^{(\ell)} & \cdots & z_{1,n_\ell}^{(\ell)} \\ \vdots & \ddots & \vdots \\ z_{n_{\ell-1},1}^{(\ell)} & \cdots & z_{n_{\ell-1},n_\ell}^{(\ell)} \end{bmatrix}}_{Z_w^{(\ell)}} \tag{7}$$

sehingga diperoleh penulisan ringkas matriks bobot tereparameterisasi pada layer ℓ:

$$W^{(\ell)} = M^{(\ell)} + \Sigma^{(\ell)} \odot Z_w^{(\ell)} \in \mathbb{R}^{n_{\ell-1} \times n_\ell} \tag{8}$$

1.2 Forward Propagation

1.2.1 Tinjau Satu Neuron

Untuk neuron ke-j pada layer ℓ ($j = 1, \ldots, n_\ell$), komputasi terhadap data ke-i ($i = 1, \ldots, N$) dirumuskan sebagai:

$$z_{i,j}^{(\ell)} = \sum_{k=1}^{n_{\ell-1}} x_{i,k}^{(\ell-1)} w_{k,j}^{(\ell)} + b_j^{(\ell)} = \sum_{k=1}^{n_{\ell-1}} x_{i,k}^{(\ell-1)} \left(\mu_{k,j}^{(\ell)} + \sigma_{k,j}^{(\ell)} z_{k,j}^{(\ell)} \right) + b_j^{(\ell)} \tag{9}$$

$$x_{i,j}^{(\ell)} = \sigma^{(\ell)} \left(z_{i,j}^{(\ell)} \right) \tag{10}$$

1.2.2 Tinjau Satu Data

Didefinisikan data ke-i, $\mathbf{x}_i^{(\ell-1)}$, bias $\mathbf{b}^{(\ell)}$, dan $\mathbf{z}_i^{(\ell)}$ pada layer ℓ:

$$\mathbf{x}_i^{(\ell-1)} = \begin{bmatrix} x_{i,1}^{(\ell-1)} & x_{i,2}^{(\ell-1)} & \cdots & x_{i,n_{\ell-1}}^{(\ell-1)} \end{bmatrix} \in \mathbb{R}^{1 \times n_{\ell-1}} \tag{11}$$

$$\mathbf{b}^{(\ell)} = \begin{bmatrix} b_1^{(\ell)} & \cdots & b_{n_\ell}^{(\ell)} \end{bmatrix} \in \mathbb{R}^{1 \times n_\ell}, \qquad \mathbf{z}_i^{(\ell)} = \begin{bmatrix} z_{i,1}^{(\ell)} & \cdots & z_{i,n_\ell}^{(\ell)} \end{bmatrix} \in \mathbb{R}^{1 \times n_\ell} \tag{12}$$

Transformasi pada layer ℓ untuk data ke-i:

$$\underbrace{\mathbf{z}_i^{(\ell)}}_{(1 \times n_\ell)} = \underbrace{\mathbf{x}_i^{(\ell-1)}}_{(1 \times n_{\ell-1})} \times \underbrace{W^{(\ell)}}_{(n_{\ell-1} \times n_\ell)} + \underbrace{\mathbf{b}^{(\ell)}}_{(1 \times n_\ell)} = \underbrace{\mathbf{x}_i^{(\ell-1)}}_{(1 \times n_{\ell-1})} \times \underbrace{\left(M^{(\ell)} + \Sigma^{(\ell)} \odot Z_w^{(\ell)} \right)}_{(n_{\ell-1} \times n_\ell)} + \underbrace{\mathbf{b}^{(\ell)}}_{(1 \times n_\ell)} \tag{13}$$

$$\underbrace{\mathbf{x}_i^{(\ell)}}_{(1 \times n_\ell)} = \sigma^{(\ell)} \left(\mathbf{z}_i^{(\ell)} \right) \tag{14}$$

1.2.3 Tinjau N Data

Diberikan matriks data input $X^{(\ell-1)} \in \mathbb{R}^{N \times n_{\ell-1}}$ dengan $X^{(0)} = X \in \mathbb{R}^{N \times D}$:

$$X^{(\ell-1)} = \begin{bmatrix} x_{1,1}^{(\ell-1)} & x_{1,2}^{(\ell-1)} & \cdots & x_{1,n_{\ell-1}}^{(\ell-1)} \\ x_{2,1}^{(\ell-1)} & x_{2,2}^{(\ell-1)} & \cdots & x_{2,n_{\ell-1}}^{(\ell-1)} \\ \vdots & \vdots & \ddots & \vdots \\ x_{N,1}^{(\ell-1)} & x_{N,2}^{(\ell-1)} & \cdots & x_{N,n_{\ell-1}}^{(\ell-1)} \end{bmatrix} \in \mathbb{R}^{N \times n_{\ell-1}} \tag{15}$$

Operasi tingkat *batch*:

$$\underbrace{Z^{(\ell)}}_{(N \times n_\ell)} = \underbrace{X^{(\ell-1)}}_{(N \times n_{\ell-1})} \times \underbrace{W^{(\ell)}}_{(n_{\ell-1} \times n_\ell)} + \underbrace{\mathbf{1}_N}_{(N \times 1)} \times \underbrace{\mathbf{b}^{(\ell)}}_{(1 \times n_\ell)} = \underbrace{X^{(\ell-1)}}_{(N \times n_{\ell-1})} \times \underbrace{\left(M^{(\ell)} + \Sigma^{(\ell)} \odot Z_w^{(\ell)} \right)}_{(n_{\ell-1} \times n_\ell)} + \underbrace{\mathbf{1}_N}_{(N \times 1)} \times \underbrace{\mathbf{b}^{(\ell)}}_{(1 \times n_\ell)} \tag{16}$$

$$\underbrace{X^{(\ell)}}_{(N \times n_\ell)} = \sigma^{(\ell)} \left(Z^{(\ell)} \right) \tag{17}$$

1.3 Backward Propagation

1.3.1 Motivasi

Pada bobot berdistribusi, parameter yang diperbarui menggunakan *Gradient Descent* dengan *learning rate* $\eta > 0$ adalah parameter distribusi $M^{(\ell)}$, $\Sigma^{(\ell)}$, serta vektor bias $\mathbf{b}^{(\ell)}$:

$$M_{n+1}^{(\ell)} = M_n^{(\ell)} - \eta \frac{\partial \mathcal{L}}{\partial M_n^{(\ell)}}, \qquad \Sigma_{n+1}^{(\ell)} = \Sigma_n^{(\ell)} - \eta \frac{\partial \mathcal{L}}{\partial \Sigma_n^{(\ell)}}, \qquad \mathbf{b}_{n+1}^{(\ell)} = \mathbf{b}_n^{(\ell)} - \eta \frac{\partial \mathcal{L}}{\partial \mathbf{b}_n^{(\ell)}} \tag{18}$$

dengan n menyatakan indeks iterasi.

1.3.2 Turunan Parameter pada Layer Output ($\ell = L$)

Diberikan $\hat{Y}$ dan fungsi objektif $\mathcal{L}$:

$$\hat{Y} = X^{(L-1)}W^{(L)} + \mathbf{1}_N \mathbf{b}^{(L)} \in \mathbb{R}^{N \times n_L}, \qquad \mathcal{L} = \frac{1}{2N} \sum_{i=1}^{N} \sum_{j=1}^{n_L} (y_{i,j} - \hat{y}_{i,j})^2 \tag{19}$$

Claim 1.1. *Turunan fungsi objektif $\mathcal{L}$ terhadap $\hat{Y} \in \mathbb{R}^{N \times n_L}$ adalah:*

$$\frac{\partial \mathcal{L}}{\partial \hat{Y}} = \frac{1}{N} \left(\hat{Y} - Y \right) \in \mathbb{R}^{N \times n_L} \tag{20}$$

Proof. Turunan fungsi skalar $\mathcal{L}$ terhadap matriks $\hat{Y}$ didefinisikan sebagai:

$$\frac{\partial \mathcal{L}}{\partial \hat{Y}} = \begin{bmatrix} \frac{\partial \mathcal{L}}{\partial \hat{y}_{1,1}} & \frac{\partial \mathcal{L}}{\partial \hat{y}_{1,2}} & \cdots & \frac{\partial \mathcal{L}}{\partial \hat{y}_{1,n_L}} \\ \frac{\partial \mathcal{L}}{\partial \hat{y}_{2,1}} & \frac{\partial \mathcal{L}}{\partial \hat{y}_{2,2}} & \cdots & \frac{\partial \mathcal{L}}{\partial \hat{y}_{2,n_L}} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial \mathcal{L}}{\partial \hat{y}_{N,1}} & \frac{\partial \mathcal{L}}{\partial \hat{y}_{N,2}} & \cdots & \frac{\partial \mathcal{L}}{\partial \hat{y}_{N,n_L}} \end{bmatrix} \in \mathbb{R}^{N \times n_L} \tag{21}$$

Dengan *chain rule*, turunan terhadap setiap elemen $\hat{y}_{i,j}$:

$$\frac{\partial \mathcal{L}}{\partial \hat{y}_{i,j}} = \frac{\partial \mathcal{L}}{\partial (y_{i,j} - \hat{y}_{i,j})} \frac{\partial (y_{i,j} - \hat{y}_{i,j})}{\partial \hat{y}_{i,j}} = \frac{1}{N}(y_{i,j} - \hat{y}_{i,j}) \cdot (-1) = \frac{1}{N}(\hat{y}_{i,j} - y_{i,j}) \tag{22}$$

Sehingga diperoleh $\frac{\partial \mathcal{L}}{\partial \hat{Y}} = \frac{1}{N} \left(\hat{Y} - Y \right)$. □

Claim 1.2. *Gradien fungsi objektif $\mathcal{L}$ terhadap matriks rata-rata bobot $M^{(L)} \in \mathbb{R}^{n_{L-1} \times n_L}$ adalah:*

$$\frac{\partial \mathcal{L}}{\partial M^{(L)}} = \frac{1}{N} \left(X^{(L-1)} \right)^T \left(\hat{Y} - Y \right) \in \mathbb{R}^{n_{L-1} \times n_L} \tag{23}$$

Proof. Dengan *chain rule* terhadap $\hat{Y} = X^{(L-1)}W^{(L)} + \mathbf{1}_N \mathbf{b}^{(L)}$:

$$\frac{\partial \mathcal{L}}{\partial W^{(L)}} = \frac{\partial \mathcal{L}}{\partial \hat{Y}} \frac{\partial \hat{Y}}{\partial W^{(L)}} = \left(X^{(L-1)} \right)^T \frac{\partial \mathcal{L}}{\partial \hat{Y}} = \frac{1}{N} \left(X^{(L-1)} \right)^T \left(\hat{Y} - Y \right) \in \mathbb{R}^{n_{L-1} \times n_L} \tag{24}$$

Untuk setiap elemen $w_{k,j}^{(L)} = \mu_{k,j}^{(L)} + \sigma_{k,j}^{(L)} z_{k,j}^{(L)}$, dengan *chain rule*:

$$\frac{\partial \mathcal{L}}{\partial \mu_{k,j}^{(L)}} = \frac{\partial \mathcal{L}}{\partial w_{k,j}^{(L)}} \frac{\partial w_{k,j}^{(L)}}{\partial \mu_{k,j}^{(L)}} = \frac{\partial \mathcal{L}}{\partial w_{k,j}^{(L)}} \cdot 1 = \frac{\partial \mathcal{L}}{\partial w_{k,j}^{(L)}} \tag{25}$$

Menyusun seluruh elemen (k, j) ke dalam bentuk matriks menghasilkan:

$$\frac{\partial \mathcal{L}}{\partial M^{(L)}} = \frac{\partial \mathcal{L}}{\partial W^{(L)}} = \frac{1}{N} \left(X^{(L-1)} \right)^T \left(\hat{Y} - Y \right) \tag{26}$$

□

Claim 1.3. *Gradien fungsi objektif $\mathcal{L}$ terhadap matriks simpangan baku bobot $\Sigma^{(L)} \in \mathbb{R}^{n_{L-1} \times n_L}$ adalah:*

$$\frac{\partial \mathcal{L}}{\partial \Sigma^{(L)}} = \left[\frac{1}{N} \left(X^{(L-1)} \right)^T \left(\hat{Y} - Y \right) \right] \odot Z_w^{(L)} \in \mathbb{R}^{n_{L-1} \times n_L} \tag{27}$$

Proof. Untuk setiap elemen $w_{k,j}^{(L)} = \mu_{k,j}^{(L)} + \sigma_{k,j}^{(L)} z_{k,j}^{(L)}$, dengan *chain rule*:

$$\frac{\partial \mathcal{L}}{\partial \sigma_{k,j}^{(L)}} = \frac{\partial \mathcal{L}}{\partial w_{k,j}^{(L)}} \frac{\partial w_{k,j}^{(L)}}{\partial \sigma_{k,j}^{(L)}} = \frac{\partial \mathcal{L}}{\partial w_{k,j}^{(L)}} \cdot z_{k,j}^{(L)} \tag{28}$$

Menyusun seluruh elemen (k, j) ke dalam bentuk matriks menghasilkan:

$$\frac{\partial \mathcal{L}}{\partial \Sigma^{(L)}} = \frac{\partial \mathcal{L}}{\partial W^{(L)}} \odot Z_w^{(L)} = \left[\frac{1}{N} \left(X^{(L-1)} \right)^T \left(\hat{Y} - Y \right) \right] \odot Z_w^{(L)} \tag{29}$$

□

Claim 1.4. *Gradien fungsi objektif $\mathcal{L}$ terhadap vektor bias $\mathbf{b}^{(L)} \in \mathbb{R}^{1 \times n_L}$ adalah:*

$$\frac{\partial \mathcal{L}}{\partial \mathbf{b}^{(L)}} = \frac{1}{N} \mathbf{1}_N^T \left(\hat{Y} - Y \right) \in \mathbb{R}^{1 \times n_L} \tag{30}$$

Proof. Dengan *chain rule* terhadap $\hat{Y} = X^{(L-1)} W^{(L)} + \mathbf{1}_N \mathbf{b}^{(L)}$:

$$\frac{\partial \mathcal{L}}{\partial \mathbf{b}^{(L)}} = \frac{\partial \mathcal{L}}{\partial \hat{Y}} \frac{\partial \hat{Y}}{\partial \mathbf{b}^{(L)}} = \mathbf{1}_N^T \frac{\partial \mathcal{L}}{\partial \hat{Y}} = \frac{1}{N} \mathbf{1}_N^T \left(\hat{Y} - Y \right) \tag{31}$$

□

1.3.3 Turunan Parameter pada Layer $\ell = L-1, \ldots, 2, 1$

Pada hidden layer $\ell \in \{L-1, \ldots, 2, 1\}$, fungsi aktivasi $\sigma^{(\ell)}$ diterapkan terhadap bentuk matriks:

$$X^{(\ell)} = \sigma^{(\ell)}\Big(X^{(\ell-1)}W^{(\ell)} + \mathbf{1}_N \mathbf{b}^{(\ell)}\Big), \qquad W^{(\ell)} = M^{(\ell)} + \Sigma^{(\ell)} \odot Z_w^{(\ell)} \in \mathbb{R}^{n_{\ell-1} \times n_\ell} \tag{32}$$

***Backward Propagation* pada Layer $\ell = L-1$:**

Backward propagation galat dari layer output ke fitur $X^{(L-1)}$:

$$\frac{\partial \mathcal{L}}{\partial X^{(L-1)}} = \frac{\partial \mathcal{L}}{\partial \hat{Y}} \frac{\partial \hat{Y}}{\partial X^{(L-1)}} = \frac{1}{N}\left(\hat{Y} - Y\right)\left(W^{(L)}\right)^T \in \mathbb{R}^{N \times n_{L-1}} \tag{33}$$

Turunan terhadap $X^{(L-2)}W^{(L-1)} + \mathbf{1}_N \mathbf{b}^{(L-1)}$:

$$\begin{aligned}\frac{\partial \mathcal{L}}{\partial \left(X^{(L-2)}W^{(L-1)} + \mathbf{1}_N \mathbf{b}^{(L-1)}\right)} &= \frac{\partial \mathcal{L}}{\partial X^{(L-1)}} \frac{\partial X^{(L-1)}}{\partial \left(X^{(L-2)}W^{(L-1)} + \mathbf{1}_N \mathbf{b}^{(L-1)}\right)} \\ &= \frac{\partial \mathcal{L}}{\partial X^{(L-1)}} \odot \sigma^{(L-1)\prime}\Big(X^{(L-2)}W^{(L-1)} + \mathbf{1}_N \mathbf{b}^{(L-1)}\Big) \in \mathbb{R}^{N \times n_{L-1}}\end{aligned} \tag{34}$$

Gradien parameter $W^{(L-1)}$, $M^{(L-1)}$, $\Sigma^{(L-1)}$, dan $\mathbf{b}^{(L-1)}$:

$$\begin{aligned}\frac{\partial \mathcal{L}}{\partial W^{(L-1)}} &= \frac{\partial \mathcal{L}}{\partial \left(X^{(L-2)}W^{(L-1)} + \mathbf{1}_N \mathbf{b}^{(L-1)}\right)} \frac{\partial \left(X^{(L-2)}W^{(L-1)} + \mathbf{1}_N \mathbf{b}^{(L-1)}\right)}{\partial W^{(L-1)}} \\ &= \left(X^{(L-2)}\right)^T \frac{\partial \mathcal{L}}{\partial \left(X^{(L-2)}W^{(L-1)} + \mathbf{1}_N \mathbf{b}^{(L-1)}\right)} \in \mathbb{R}^{n_{L-2} \times n_{L-1}}\end{aligned} \tag{35}$$

$$\begin{aligned}\frac{\partial \mathcal{L}}{\partial M^{(L-1)}} &= \frac{\partial \mathcal{L}}{\partial \left(X^{(L-2)}W^{(L-1)} + \mathbf{1}_N \mathbf{b}^{(L-1)}\right)} \frac{\partial \left(X^{(L-2)}W^{(L-1)} + \mathbf{1}_N \mathbf{b}^{(L-1)}\right)}{\partial W^{(L-1)}} \frac{\partial W^{(L-1)}}{\partial M^{(L-1)}} = \frac{\partial \mathcal{L}}{\partial W^{(L-1)}} \frac{\partial W^{(L-1)}}{\partial M^{(L-1)}} \\ &= \frac{\partial \mathcal{L}}{\partial W^{(L-1)}} = \left(X^{(L-2)}\right)^T \frac{\partial \mathcal{L}}{\partial \left(X^{(L-2)}W^{(L-1)} + \mathbf{1}_N \mathbf{b}^{(L-1)}\right)} \in \mathbb{R}^{n_{L-2} \times n_{L-1}}\end{aligned} \tag{36}$$

$$\begin{aligned}\frac{\partial \mathcal{L}}{\partial \Sigma^{(L-1)}} &= \frac{\partial \mathcal{L}}{\partial \left(X^{(L-2)}W^{(L-1)} + \mathbf{1}_N \mathbf{b}^{(L-1)}\right)} \frac{\partial \left(X^{(L-2)}W^{(L-1)} + \mathbf{1}_N \mathbf{b}^{(L-1)}\right)}{\partial W^{(L-1)}} \frac{\partial W^{(L-1)}}{\partial \Sigma^{(L-1)}} = \frac{\partial \mathcal{L}}{\partial W^{(L-1)}} \frac{\partial W^{(L-1)}}{\partial \Sigma^{(L-1)}} \\ &= \frac{\partial \mathcal{L}}{\partial W^{(L-1)}} \odot Z_w^{(L-1)} = \left[\left(X^{(L-2)}\right)^T \frac{\partial \mathcal{L}}{\partial \left(X^{(L-2)}W^{(L-1)} + \mathbf{1}_N \mathbf{b}^{(L-1)}\right)}\right] \odot Z_w^{(L-1)} \in \mathbb{R}^{n_{L-2} \times n_{L-1}}\end{aligned} \tag{37}$$

$$\begin{aligned}\frac{\partial \mathcal{L}}{\partial \mathbf{b}^{(L-1)}} &= \frac{\partial \mathcal{L}}{\partial \left(X^{(L-2)}W^{(L-1)} + \mathbf{1}_N \mathbf{b}^{(L-1)}\right)} \frac{\partial \left(X^{(L-2)}W^{(L-1)} + \mathbf{1}_N \mathbf{b}^{(L-1)}\right)}{\partial \mathbf{b}^{(L-1)}} \\ &= \mathbf{1}_N^T \frac{\partial \mathcal{L}}{\partial \left(X^{(L-2)}W^{(L-1)} + \mathbf{1}_N \mathbf{b}^{(L-1)}\right)} \in \mathbb{R}^{1 \times n_{L-1}}\end{aligned} \tag{38}$$

Backward propagation ke fitur layer $L-2$:

$$\begin{aligned}\frac{\partial \mathcal{L}}{\partial X^{(L-2)}} &= \frac{\partial \mathcal{L}}{\partial \left(X^{(L-2)}W^{(L-1)} + \mathbf{1}_N \mathbf{b}^{(L-1)}\right)} \frac{\partial \left(X^{(L-2)}W^{(L-1)} + \mathbf{1}_N \mathbf{b}^{(L-1)}\right)}{\partial X^{(L-2)}} \\ &= \frac{\partial \mathcal{L}}{\partial \left(X^{(L-2)}W^{(L-1)} + \mathbf{1}_N \mathbf{b}^{(L-1)}\right)} \left(W^{(L-1)}\right)^T \in \mathbb{R}^{N \times n_{L-2}}\end{aligned} \tag{39}$$

***Backward Propagation* pada Layer $\ell = L-2$:**

Turunan terhadap $X^{(L-3)}W^{(L-2)} + \mathbf{1}_N \mathbf{b}^{(L-2)}$:

$$\begin{aligned}\frac{\partial \mathcal{L}}{\partial \left(X^{(L-3)}W^{(L-2)} + \mathbf{1}_N \mathbf{b}^{(L-2)}\right)} &= \frac{\partial \mathcal{L}}{\partial X^{(L-2)}} \frac{\partial X^{(L-2)}}{\partial \left(X^{(L-3)}W^{(L-2)} + \mathbf{1}_N \mathbf{b}^{(L-2)}\right)} \\ &= \frac{\partial \mathcal{L}}{\partial X^{(L-2)}} \odot \sigma^{(L-2)\prime}\Big(X^{(L-3)}W^{(L-2)} + \mathbf{1}_N \mathbf{b}^{(L-2)}\Big) \in \mathbb{R}^{N \times n_{L-2}}\end{aligned} \tag{40}$$

Gradien parameter $W^{(L-2)}$, $M^{(L-2)}$, $\Sigma^{(L-2)}$, dan $\mathbf{b}^{(L-2)}$:

$$\frac{\partial \mathcal{L}}{\partial W^{(L-2)}} = \frac{\partial \mathcal{L}}{\partial \left(X^{(L-3)}W^{(L-2)} + \mathbf{1}_N \mathbf{b}^{(L-2)}\right)} \frac{\partial \left(X^{(L-3)}W^{(L-2)} + \mathbf{1}_N \mathbf{b}^{(L-2)}\right)}{\partial W^{(L-2)}}$$
$$= \left(X^{(L-3)}\right)^T \frac{\partial \mathcal{L}}{\partial \left(X^{(L-3)}W^{(L-2)} + \mathbf{1}_N \mathbf{b}^{(L-2)}\right)} \in \mathbb{R}^{n_{L-3} \times n_{L-2}} \tag{41}$$

$$\frac{\partial \mathcal{L}}{\partial M^{(L-2)}} = \frac{\partial \mathcal{L}}{\partial \left(X^{(L-3)}W^{(L-2)} + \mathbf{1}_N \mathbf{b}^{(L-2)}\right)} \frac{\partial \left(X^{(L-3)}W^{(L-2)} + \mathbf{1}_N \mathbf{b}^{(L-2)}\right)}{\partial W^{(L-2)}} \frac{\partial W^{(L-2)}}{\partial M^{(L-2)}} = \frac{\partial \mathcal{L}}{\partial W^{(L-2)}} \frac{\partial W^{(L-2)}}{\partial M^{(L-2)}}$$
$$= \frac{\partial \mathcal{L}}{\partial W^{(L-2)}} = \left(X^{(L-3)}\right)^T \frac{\partial \mathcal{L}}{\partial \left(X^{(L-3)}W^{(L-2)} + \mathbf{1}_N \mathbf{b}^{(L-2)}\right)} \in \mathbb{R}^{n_{L-3} \times n_{L-2}} \tag{42}$$

$$\frac{\partial \mathcal{L}}{\partial \Sigma^{(L-2)}} = \frac{\partial \mathcal{L}}{\partial \left(X^{(L-3)}W^{(L-2)} + \mathbf{1}_N \mathbf{b}^{(L-2)}\right)} \frac{\partial \left(X^{(L-3)}W^{(L-2)} + \mathbf{1}_N \mathbf{b}^{(L-2)}\right)}{\partial W^{(L-2)}} \frac{\partial W^{(L-2)}}{\partial \Sigma^{(L-2)}} = \frac{\partial \mathcal{L}}{\partial W^{(L-2)}} \frac{\partial W^{(L-2)}}{\partial \Sigma^{(L-2)}}$$
$$= \frac{\partial \mathcal{L}}{\partial W^{(L-2)}} \odot Z_w^{(L-2)} = \left[\left(X^{(L-3)}\right)^T \frac{\partial \mathcal{L}}{\partial \left(X^{(L-3)}W^{(L-2)} + \mathbf{1}_N \mathbf{b}^{(L-2)}\right)}\right] \odot Z_w^{(L-2)} \in \mathbb{R}^{n_{L-3} \times n_{L-2}} \tag{43}$$

$$\frac{\partial \mathcal{L}}{\partial \mathbf{b}^{(L-2)}} = \frac{\partial \mathcal{L}}{\partial \left(X^{(L-3)}W^{(L-2)} + \mathbf{1}_N \mathbf{b}^{(L-2)}\right)} \frac{\partial \left(X^{(L-3)}W^{(L-2)} + \mathbf{1}_N \mathbf{b}^{(L-2)}\right)}{\partial \mathbf{b}^{(L-2)}}$$
$$= \mathbf{1}_N^T \frac{\partial \mathcal{L}}{\partial \left(X^{(L-3)}W^{(L-2)} + \mathbf{1}_N \mathbf{b}^{(L-2)}\right)} \in \mathbb{R}^{1 \times n_{L-2}} \tag{44}$$

Backward propagation ke fitur layer $L-3$:

$$\frac{\partial \mathcal{L}}{\partial X^{(L-3)}} = \frac{\partial \mathcal{L}}{\partial \left(X^{(L-3)}W^{(L-2)} + \mathbf{1}_N \mathbf{b}^{(L-2)}\right)} \frac{\partial \left(X^{(L-3)}W^{(L-2)} + \mathbf{1}_N \mathbf{b}^{(L-2)}\right)}{\partial X^{(L-3)}}$$
$$= \frac{\partial \mathcal{L}}{\partial \left(X^{(L-3)}W^{(L-2)} + \mathbf{1}_N \mathbf{b}^{(L-2)}\right)} \left(W^{(L-2)}\right)^T \in \mathbb{R}^{N \times n_{L-3}} \tag{45}$$

***Backward Propagation* Umum pada Hidden Layer ($\ell = L-1, \ldots, 2, 1$):**

Secara umum, untuk setiap *hidden layer* $\ell \in \{L-1, \ldots, 2, 1\}$:

$$\frac{\partial \mathcal{L}}{\partial \left(X^{(\ell-1)}W^{(\ell)} + \mathbf{1}_N \mathbf{b}^{(\ell)}\right)} = \frac{\partial \mathcal{L}}{\partial X^{(\ell)}} \frac{\partial X^{(\ell)}}{\partial \left(X^{(\ell-1)}W^{(\ell)} + \mathbf{1}_N \mathbf{b}^{(\ell)}\right)}$$

$$= \frac{\partial \mathcal{L}}{\partial X^{(\ell)}} \odot \sigma^{(\ell)\prime}\left(X^{(\ell-1)}W^{(\ell)} + \mathbf{1}_N \mathbf{b}^{(\ell)}\right) \in \mathbb{R}^{N \times n_\ell} \tag{46}$$

$$\frac{\partial \mathcal{L}}{\partial W^{(\ell)}} = \frac{\partial \mathcal{L}}{\partial \left(X^{(\ell-1)}W^{(\ell)} + \mathbf{1}_N \mathbf{b}^{(\ell)}\right)} \frac{\partial \left(X^{(\ell-1)}W^{(\ell)} + \mathbf{1}_N \mathbf{b}^{(\ell)}\right)}{\partial W^{(\ell)}}$$

$$= \left(X^{(\ell-1)}\right)^T \frac{\partial \mathcal{L}}{\partial \left(X^{(\ell-1)}W^{(\ell)} + \mathbf{1}_N \mathbf{b}^{(\ell)}\right)} \in \mathbb{R}^{n_{\ell-1} \times n_\ell} \tag{47}$$

$$\frac{\partial \mathcal{L}}{\partial M^{(\ell)}} = \frac{\partial \mathcal{L}}{\partial \left(X^{(\ell-1)}W^{(\ell)} + \mathbf{1}_N \mathbf{b}^{(\ell)}\right)} \frac{\partial \left(X^{(\ell-1)}W^{(\ell)} + \mathbf{1}_N \mathbf{b}^{(\ell)}\right)}{\partial W^{(\ell)}} \frac{\partial W^{(\ell)}}{\partial M^{(\ell)}} = \frac{\partial \mathcal{L}}{\partial W^{(\ell)}} \frac{\partial W^{(\ell)}}{\partial M^{(\ell)}}$$

$$= \frac{\partial \mathcal{L}}{\partial W^{(\ell)}} = \left(X^{(\ell-1)}\right)^T \frac{\partial \mathcal{L}}{\partial \left(X^{(\ell-1)}W^{(\ell)} + \mathbf{1}_N \mathbf{b}^{(\ell)}\right)} \in \mathbb{R}^{n_{\ell-1} \times n_\ell} \tag{48}$$

$$\frac{\partial \mathcal{L}}{\partial \Sigma^{(\ell)}} = \frac{\partial \mathcal{L}}{\partial \left(X^{(\ell-1)}W^{(\ell)} + \mathbf{1}_N \mathbf{b}^{(\ell)}\right)} \frac{\partial \left(X^{(\ell-1)}W^{(\ell)} + \mathbf{1}_N \mathbf{b}^{(\ell)}\right)}{\partial W^{(\ell)}} \frac{\partial W^{(\ell)}}{\partial \Sigma^{(\ell)}} = \frac{\partial \mathcal{L}}{\partial W^{(\ell)}} \frac{\partial W^{(\ell)}}{\partial \Sigma^{(\ell)}}$$

$$= \frac{\partial \mathcal{L}}{\partial W^{(\ell)}} \odot Z_w^{(\ell)} = \left[\left(X^{(\ell-1)}\right)^T \frac{\partial \mathcal{L}}{\partial \left(X^{(\ell-1)}W^{(\ell)} + \mathbf{1}_N \mathbf{b}^{(\ell)}\right)}\right] \odot Z_w^{(\ell)} \in \mathbb{R}^{n_{\ell-1} \times n_\ell} \tag{49}$$

$$\frac{\partial \mathcal{L}}{\partial \mathbf{b}^{(\ell)}} = \frac{\partial \mathcal{L}}{\partial \left(X^{(\ell-1)}W^{(\ell)} + \mathbf{1}_N \mathbf{b}^{(\ell)}\right)} \frac{\partial \left(X^{(\ell-1)}W^{(\ell)} + \mathbf{1}_N \mathbf{b}^{(\ell)}\right)}{\partial \mathbf{b}^{(\ell)}}$$

$$= \mathbf{1}_N^T \frac{\partial \mathcal{L}}{\partial \left(X^{(\ell-1)}W^{(\ell)} + \mathbf{1}_N \mathbf{b}^{(\ell)}\right)} \in \mathbb{R}^{1 \times n_\ell} \tag{50}$$

$$\frac{\partial \mathcal{L}}{\partial X^{(\ell-1)}} = \frac{\partial \mathcal{L}}{\partial \left(X^{(\ell-1)}W^{(\ell)} + \mathbf{1}_N \mathbf{b}^{(\ell)}\right)} \frac{\partial \left(X^{(\ell-1)}W^{(\ell)} + \mathbf{1}_N \mathbf{b}^{(\ell)}\right)}{\partial X^{(\ell-1)}}$$

$$= \frac{\partial \mathcal{L}}{\partial \left(X^{(\ell-1)}W^{(\ell)} + \mathbf{1}_N \mathbf{b}^{(\ell)}\right)} \left(W^{(\ell)}\right)^T \in \mathbb{R}^{N \times n_{\ell-1}} \tag{51}$$

dengan kondisi batas input awal $X^{(0)} = X \in \mathbb{R}^{N \times D}$.